

미적 2차과제

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9.4

$$1. \quad x = r \cos \theta, y = r \sin \theta \quad r' = \cos \theta - \sin \theta, \frac{dy}{dx} = \frac{r' \sin \theta + r \cos \theta}{r' \cos \theta - r \sin \theta} = -1$$

$$\theta = \frac{\pi}{4} \quad x=1, y=1 \quad y' = -(2-1)+1 = -x+2$$

$$2. \quad \int_a^b \pi r^2 d\theta \rightarrow \int_0^{2\pi} \pi \cdot \cos^2 \theta d\theta = \int_0^{2\pi} \pi \cdot \frac{2(\cos 2\theta + 1)}{2} d\theta$$

$$= \int_0^{2\pi} \pi \cdot \cos^2 \theta d\theta = \int_0^{2\pi} \pi \cdot \frac{2(\cos 2\theta + 1)}{2} d\theta = \frac{\pi}{2} \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{2\pi}$$

$$= \frac{\pi}{2} \cdot \left(\frac{2\pi}{1} \right) = \pi \rightarrow k > \frac{25}{\pi} \quad (k=8)$$

3.

$$4. \quad 1 - (\cos \theta) + 1 + \cos \theta = 2 \quad 0 \leq \theta \leq \frac{\pi}{2}$$

$$\frac{1}{2} \int_a^b r^2 d\theta \rightarrow 4\pi \int_0^{\frac{\pi}{2}} (1 - \cos \theta)^2 d\theta = 2 \int_0^{\frac{\pi}{2}} (1 - 2\cos \theta + \cos^2 \theta) d\theta = 2 \int_0^{\frac{\pi}{2}} (1 - 2\cos \theta + \frac{1}{2} + \frac{1}{2} \cos 2\theta) d\theta$$

$$= 2 \left[\frac{3}{2} \theta - 2\sin \theta + \frac{1}{4} \sin 2\theta \right]_0^{\frac{\pi}{2}} = 2 \left(\frac{3\pi}{4} - 2 \right) = \frac{3}{2} \pi - 4$$

$$5. \quad x = r \cos \theta = (1 + \cos \theta) \cos \theta, y = r \sin \theta = (1 + \cos \theta) \sin \theta$$

$$\Rightarrow (1 + \cos \theta) \cos \theta = (1 + \cos \theta) \sin \theta \Rightarrow \theta = \frac{\pi}{4} \text{ or } \frac{3\pi}{4} \text{ or } \pi$$

$$\Rightarrow r = 1 + \frac{\sqrt{2}}{2}, 1 - \frac{\sqrt{2}}{2}, 0 \quad (x, y) = \left(\frac{\sqrt{2}+1}{2}, \frac{\sqrt{2}+1}{2} \right), \left(\frac{1-\sqrt{2}}{2}, \frac{1-\sqrt{2}}{2} \right), (0, 0)$$

$$6. \quad L = 2 \int_0^{\frac{\pi}{2}} \sqrt{1 + 2\cos \theta \sin \theta + \sin^2 \theta} d\theta = 2 \int_0^{\frac{\pi}{2}} \sqrt{2(1 + \cos \theta)} d\theta = 2 \int_0^{\frac{\pi}{2}} 2 \cos \frac{\theta}{2} d\theta = 4 \int_0^{\frac{\pi}{2}} \cos \frac{\theta}{2} d\theta$$

$$= \left[8 \sin \frac{\theta}{2} \right]_0^{\frac{\pi}{2}} = 8$$

11.1

$$1. (a): r = \sqrt{(-1-6)^2 + (2-(-2))^2 + (1-2)^2} = \sqrt{69} \rightarrow (x+1)^2 + (y-2)^2 + (z-1)^2 = 69$$

$$b): x=0 \quad (y-2)^2 + (z-1)^2 = 68$$

$$c): (x-4)^2 + (y+1)^2 + (z+3)^2 = 25 \rightarrow \frac{3}{2} \pi, (4, -1, -3) \text{ at } \frac{3}{2} \pi$$

$$0 < x < 2 + |a|, (b): z=0, x^2 + y^2 = 4 \quad x^2 + y^2 \leq 4 \text{ at } z=0$$

$$c): x^2 + y^2 + z^2 \leq 4, z \geq 0 \text{ at } \theta = \frac{\pi}{2}$$

$$11.2 \quad (2): 0 \quad (3): AC \quad (4): AB$$

$$a) \cdot \langle -3, -(-5), 3, (-1) \rangle = \langle -2, 4 \rangle$$

$$b) \cdot \langle 0, -3, 5, 0, 0, (-2) \rangle = \langle -3, 5, 2 \rangle$$

$$11) \quad a+b = 6i+3j, 4a+2b = 6i+4j, |a|=5, |a-b|=13$$

$$a+b = \langle 6, 3, -2 \rangle, 4a+2b = \langle 20, -12, 0 \rangle, |a|=5, |a-b|=9$$

$$u = \frac{\langle 6, 3, -2 \rangle}{|\langle 6, 3, -2 \rangle|} = \left\langle \frac{6}{7}, \frac{3}{7}, -\frac{2}{7} \right\rangle$$

$$f = \frac{(-6i-2j)+3k}{|(-6i-2j)+3k|} = \frac{(-6i-2j)+3k}{7} = \left\langle -\frac{6}{7}i - \frac{2}{7}j + \frac{3}{7}k \right\rangle$$

11.3

$$1. \quad a \cdot b = \frac{1}{2} - \frac{1}{2} = 0$$

$$(2): \langle 1, 2, 3 \rangle \cdot \langle 1, -1, 5 \rangle = 1 - 2 + 15 = 14$$

$$(3): a \cdot b = |a||b|\cos \theta = 5 \times 8 \times \cos \frac{3}{4}\pi = -20\sqrt{2}$$

$$2. \quad (1): \cos \theta = \frac{a \cdot b}{|a||b|} = \frac{5}{5\sqrt{5}} = \frac{1}{\sqrt{5}} \rightarrow \theta = \cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$$

$$(2): \cos \theta = \frac{a \cdot b}{|a||b|} = \frac{8-1}{\sqrt{5}\sqrt{26}} = \frac{7}{\sqrt{130}} \rightarrow \theta = \cos^{-1}\left(\frac{7}{\sqrt{130}}\right)$$

$$3. \quad \vec{PQ} = \langle -2, 3 \rangle, \vec{QR} = \langle 3, 1 \rangle, \vec{PR} = \langle 1, 4 \rangle$$

$$\cos \angle P = \frac{|\vec{PQ} \cdot \vec{PR}|}{|\vec{PQ}||\vec{PR}|} = \frac{1101}{13\sqrt{17}} = \frac{10}{\sqrt{21}} \quad \angle P = \cos^{-1}\left(\frac{10}{\sqrt{21}}\right)$$

$$\cos \angle Q = \frac{|\vec{QR} \cdot \vec{QP}|}{|\vec{QR}||\vec{QP}|} = \frac{10}{\sqrt{10}\sqrt{13}} = \frac{3}{\sqrt{130}} \quad \angle Q = \cos^{-1}\left(\frac{3}{\sqrt{130}}\right)$$

$$\cos \angle R = \frac{|\vec{RP} \cdot \vec{RQ}|}{|\vec{RP}||\vec{RQ}|} = \frac{1}{\sqrt{17}\sqrt{10}} = \frac{1}{\sqrt{170}} \quad \angle R = \cos^{-1}\left(\frac{1}{\sqrt{170}}\right)$$

$$4. \quad \langle 3, 2, x \rangle \cdot \langle 2x, 4, x \rangle = 0 \quad 6x + 8 + x^2 = 0 \quad x = -2 \text{ or } x = -4$$

$$6. \quad a \cdot b = 3 - 2 - 3 = -1 \quad \cos \theta = \frac{a \cdot b}{|a||b|} = \frac{-1}{\sqrt{6}} = \frac{1}{\sqrt{6}} \quad \theta = \cos^{-1}\left(\frac{1}{\sqrt{6}}\right)$$

$$\text{proj}_a b = \frac{a \cdot b}{|a|^2} a = \frac{-1}{6} \langle 1, 1, -2 \rangle = \left\langle -\frac{1}{6}, -\frac{1}{6}, \frac{1}{3} \right\rangle$$

$$(2): \begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & 1 \\ 0 & 2 & -4 \\ -1 & 3 & 1 \end{pmatrix} \quad 14 \neq 4 + 2 \quad \text{Ans } b = \langle 14, 4, 2 \rangle$$

$$(2): \begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & 1 \\ 2 & 3 & 0 \\ 1 & 0 & 5 \end{pmatrix} \quad \text{Ans } b = \langle 15, -10, -3 \rangle$$

$$9. \text{ 2. } \begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & 2 & 2 \\ -4 & -3 & 2 \end{pmatrix} \quad \text{Ans } b = \langle 10, -10, 5 \rangle \rightarrow 3 \times \frac{\langle 10, -10, 5 \rangle}{\langle 10, -10, 5 \rangle} = \langle 2, -2, 1 \rangle$$

$$3. \quad U \cdot V = -2 + 1 + 1 = 0, \quad U \cdot W = -3 + 3 = 0, \quad V \cdot W = -3 + 3 = 0$$

$$V = |U||V||W| = \sqrt{36} \cdot 8 = 18$$

$$5. (1): \vec{AB} = \langle -2, 1, -1 \rangle, \vec{AC} = \langle -2, 3, 1 \rangle \quad \vec{AB} \cdot \vec{AC} = 4 + 3 - 1 = 6$$

$$(2): \vec{AB} \cdot \vec{AC} = \begin{pmatrix} 1 & -1 & 1 \\ -2 & 1 & -1 \\ -2 & 3 & 1 \end{pmatrix} = \langle 4, 4, -4 \rangle \rightarrow \frac{1}{3} |\vec{AB} \cdot \vec{AC}| = 2\sqrt{3}$$

$$5. (1): \vec{AB} = \langle -4, 2, 4 \rangle, \vec{AE} = \langle -3, 4, 1 \rangle, \vec{AD} = \langle 2, 1, -2 \rangle$$

$$\vec{AB} \cdot (\vec{AD} \times \vec{AE}) = \begin{pmatrix} 1 & -2 & 8 \\ -3 & 4 & 1 \end{pmatrix} \cdot \langle -4, 2, 4 \rangle = -36 + 8 + 44 = 16$$

$$(2): \vec{AB} \cdot \vec{AD} = \begin{pmatrix} -4 & 2 & 4 \\ 2 & 1 & -2 \end{pmatrix} = \langle -8, 0, 8 \rangle \quad \frac{1}{20} |0| = 8\sqrt{2} \quad h = \frac{16}{8\sqrt{2}} = \sqrt{2}$$